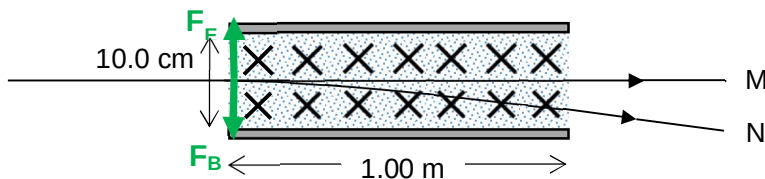


- 22 In the region of dimensions 10.0 cm by 1.00 m between two charged plates with a p.d. of 100 V, there is a uniform magnetic field of strength 100 mT directed into the page.



M and N are the paths made by electrons of different speeds.

Which of the following correctly describes the speed of the electron that made N?

- A slightly smaller than $1.00 \times 10^4 \text{ m s}^{-1}$ M: $F_B = F_E \rightarrow B q v = q E$
- B** slightly greater than $1.00 \times 10^4 \text{ m s}^{-1}$ $v = E / B = V / d B = 1.00 \times 10^4 \text{ m s}^{-1}$
- C slightly smaller than $1.00 \times 10^5 \text{ m s}^{-1}$ N: $F_B > F_E$
- D slightly greater than $1.00 \times 10^5 \text{ m s}^{-1}$ $v > 1.00 \times 10^4 \text{ m s}^{-1}$

23 A circular loop of wire with radius of 0.20 m is placed in a uniform magnetic field of 1.2 T that is